

Reliable Data at Scale: ETL vs. Unified Platforms in Modern SRE

VENKATA NAGENDRA KUMAR KUNDAVARAM

IT DEVELOPMENT MANAGER AT GOODWILL EASTER

SEALS MINNESOTA

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Agenda Overview

INTRODUCTION

This section provides an essential context for understanding the challenges and opportunities presented by data in modern SRE environments.

TRADITIONAL ETL

Explore the core components and functionalities of traditional ETL tools, emphasizing their strengths and limitations within the SRE landscape.

UNIFIED PLATFORMS

Gain insights into unified data platforms, focusing on their architecture, capabilities, and relevance for enhancing data reliability and operational efficiency.

The Growing Challenge of Data Scalability

INCREASING DATA COMPLEXITY

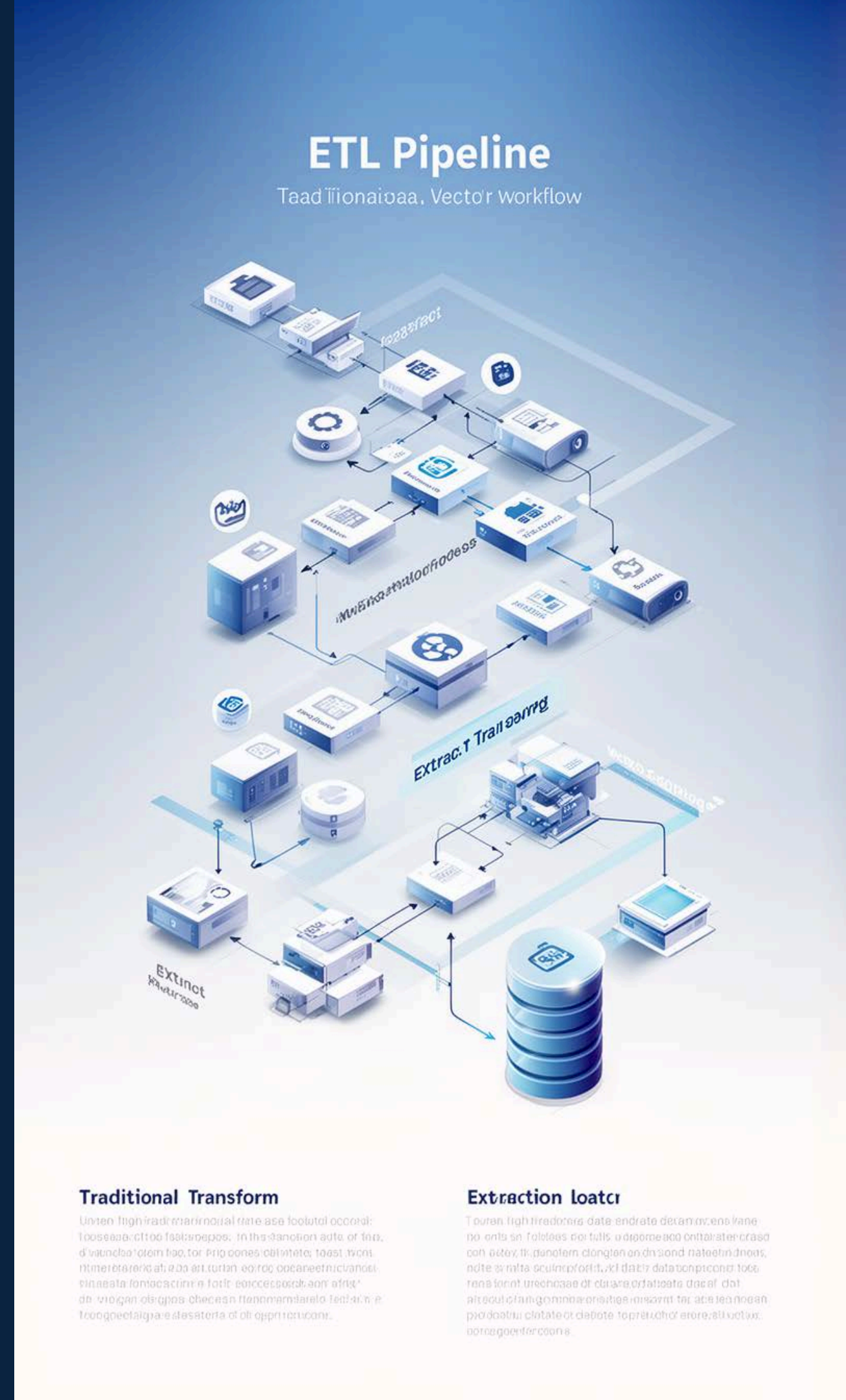
As enterprise data volume surges, SRE teams face **new challenges** in ensuring reliability and operational efficiency in managing diverse data types.



Understanding Traditional ETL Tools in SRE

BATCH-ORIENTED WORKFLOWS

Traditional ETL tools focus on **structured workflows** to extract, transform, and load data, often leading to predictable but rigid processing.



Understanding Unified Data Platforms

INTEGRATED, DISTRIBUTED ARCHITECTURE

Unified data platforms provide a **robust framework** for managing diverse data types, enhancing observability and governance while enabling seamless data ingestion. These platforms harness the power of integrated, distributed architecture to ensure that data is accessible, reliable, and secure across various environments. By centralizing data management, they facilitate better collaboration between teams, streamline data workflows, and support advanced analytics capabilities. This holistic approach empowers organizations to make data-driven decisions with greater agility and precision, ultimately driving innovation and competitive advantage in today's fast-paced digital landscape.

Scalability: ETL vs. Unified Platforms

ETL VERTICAL SCALING

Traditional ETL tools rely on vertical scaling, which often leads to **bottlenecks** as data loads increase, affecting performance and reliability.

UNIFIED PLATFORMS

Unified platforms employ horizontal scaling, enabling real-time processing and **elastic resource allocation**, which facilitates handling larger data volumes efficiently.

Flexibility in Data Handling

ETL SCHEMA OPTIMIZATION

Traditional ETL tools excel at processing **structured data**, relying on predefined schemas that ensure consistency and predictability in workflows.

UNIFIED PLATFORM SUPPORT

Unified platforms accommodate a range of **data types**, enabling seamless integration of structured, semi-structured, and unstructured data for enhanced adaptability.

Operational Cost Efficiency

ETL COST DYNAMICS

Traditional ETL tools offer **lower initial costs**, but maintenance expenses can escalate significantly as data volumes and complexity increase over time.

UNIFIED PLATFORMS

Unified data platforms require a **higher upfront investment** but can lead to long-term savings through reduced operational costs and improved efficiency.

Practical Decision Framework

SCALE

Assessing the **data volume** and growth patterns is crucial for determining the best platform to support scalability and performance requirements.

RELIABILITY

Establishing reliability targets helps ensure that the chosen data architecture meets the **availability** needs of the organization and its stakeholders.

INTEGRATION

Understanding existing systems and integration needs guides the selection process, ensuring that the platform aligns with current **operational workflows** and data practices.

Scale & Performance

CHOOSING THE RIGHT APPROACH

Batch ETL is suitable for predictable workloads, while unified platforms are essential when real-time data processing and scalability become critical. Consider your unique use case. For immediate insights from large data volumes, streaming is beneficial for its continuous processing and near-instant results. If cost-efficiency with some latency is acceptable, batch processing is reliable.

Integrating machine learning models into your data pipeline enhances decision-making and analytical insights. Ensure your platform supports such integrations for more intelligent data handling.

Evaluate your infrastructure, growth plans, and data needs to choose the most effective approach, balancing speed, accuracy, and cost with your business objectives.

Reliability



FAULT TOLERANCE AND MONITORING

Choosing the right platform directly impacts **fault tolerance** and incident detection capabilities, enhancing overall operational reliability for SRE teams.

Cost Considerations

UPFRONT INVESTMENT VS. LONG-TERM SAVINGS

Balancing initial costs with potential operational savings is crucial. **Resource availability** and required skill sets also impact overall efficiency and sustainability. Assessing the total cost of ownership is crucial. By analyzing immediate expenses and future benefits, organizations can make informed, strategic decisions. A thorough cost-benefit analysis can reveal how initial investments lead to savings, such as through energy-efficient technologies.

Additionally, solutions should be scalable and adaptable to evolving needs. Investing in staff training enhances skills, reducing the need for external hires and promoting continuous improvement. This positions businesses for sustainable growth, ensuring both immediate and long-term financial health.

Scenario Application Framework

COMPANY PROFILE

A mid-sized e-commerce firm faces **significant data challenges** as transaction volumes surge, impacting their ability to maintain operational reliability.

EVALUATION SUMMARY

Comparing ETL tools and unified platforms reveals that adopting a unified approach enhances scalability and flexibility essential for the company's growth.

Key Takeaways

ETL TOOLS

ETL tools remain effective for stable, batch workloads, allowing organizations to manage structured data efficiently with predictable performance metrics.

UNIFIED PLATFORMS

Unified platforms support scalability and flexibility, enabling organizations to adapt to diverse data types and real-time processing demands in complex environments.

DECISION FRAMEWORKS

Practical **decision frameworks** guide SRE teams in selecting data solutions that align with their operational goals, reliability targets, and cost constraints.

Thank You for Your Attention

Please feel free to reach out with any further inquiries or discussion. Let's collaborate on enhancing our data strategies and solutions.